Reg. No.	R	A	2	2	1	1	0	3	7	0	1	0	0	0	2

B.Tech/ M.Tech (Integrated) DEGREE EXAMINATION, MAY 2023

First/ Second Semester

21GNH101J – PHILOSOPHY OF ENGINEERING

(For the candidates admitted from the academic year 2022-2023)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours

Max. Marks: 75

	Marks	BL	CO	PO

PART – A (20 × 1 = 20Marks)

Answer **ALL** Questions

1. The earliest Civil Engineer is known by name — 1 | 1 | 1 | 1
   (A) Imhotep (B) Babylon
   (C) Teotihulan (D) Saqqara

2. The term engine is derived from — 1 | 1 | 1 | 1
   (A) Greek (B) Latin
   (C) English (D) Arabic

3. Modern ERA is between — 1 | 1 | 1 | 1
   (A) 1114 and 1200 (B) 1104 and 1200
   (C) 1174 and 1200 (D) 1074 and 1200

4. The first steam engine was built in — 1 | 1 | 1 | 1
   (A) 1947 (B) 1698
   (C) 1998 (D) 1647

5. _____ belongs to the major branch of metaphysics. — 1 | 1 | 2 | 1
   (A) Physiology (B) Biology
   (C) Ontology (D) Sociology

6. _____ ontologies describes concepts depending both on a particular domain and task — 1 | 1 | 2 | 1
   (A) Reference (B) Application
   (C) Theoretical (D) Metaphysical

7. What happens in the first stage of product life cycle? — 1 | 1 | 2 | 1
   (A) Product development (B) Product growth
   (C) Product maturity (D) Product decline

8. When a product is in its decline stage of its product life cycle, the average selling price will be — 1 | 1 | 2 | 1
   (A) Higher (B) Average
   (C) As decided by manufacturer (D) Lower

9. _____ is the study of the natural world as it is
(A) Engineering
(B) Social
(C) Technology
(D) Science

10. Design as _____ is related to the conceptualization stages of making new products
(A) Activity
(B) Planning
(C) Execution
(D) Epistemology

11. Holland's theory describes _____ number of basic personality types.
(A) 2
(B) 4
(C) 6
(D) 8

12. _____ focus on ideas
(A) Creators
(B) Helpers
(C) Thinkers
(D) Doers

13. The _____ test your hypothesis by doing an experiment
(A) Scientific method
(B) Addie model
(C) Holland code
(D) Engineering method

14. _____ number of parts are involved in evaluation phase.
(A) 2
(B) 4
(C) 6
(D) 8

15. The final stage of engineering design process is
(A) Define problem
(B) Research ideas
(C) Communicate results
(D) Establish constraints

16. The prototype creation is involved in _____ phase of addie model.
(A) Evaluation phase
(B) Implementation phase
(C) Development phase
(D) Design phase

17. _____ is the most discussed aspect of sustainability.
(A) Environment
(B) Economic
(C) Ethical
(D) Equity

18. The American Association of Engineering Societies was established in
(A) 1997
(B) 1979
(C) 1897
(D) 1889

19. Development which meets the needs of current generations without compromising the ability of future generations to meet their needs
(A) Sustainability
(B) Diversity
(C) Equity
(D) Integrity

20. _____ are involved in planning and managing projects
(A) Actors
(B) Teachers
(C) Doctors
(D) Engineers

PART – B (4 × 10 = 40 Marks)

Answer **ANY FOUR** Questions

		Marks	BL	CO	PO
21.	Explain briefly on non-motivated functions.	10	1	1	1
22.	Explain reference ontology with regard to application ontology.	10	1	2	1
23.	What are the four dimensions of engineering? Write short notes on these dimensions.	10	1	3	1
24.	Differentiate between scientific method and engineering design.	10	1	2	1
25.	What do you mean by CDIO engineers in industry? Explain in detail.	10	1	4	1
26.	Mention the professional organizations available for engineers with a detailed note.	10	1	5	1

PART – C (1 × 15 = 15 Marks)

Answer **ANY ONE** Question

		Marks	BL	CO	PO
27.	What do you mean by product life cycle? Explain its stages in detail.	15	1	2	
28.	Give detailed notes on RAISEC model.	15	1	2	

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